Abhijit Kumar Ghosh
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GS Google Scholar | **RG** ResearchGate | **ORCID** ORCID | **in** LinkedIn | **GitHub** GitHub

Research Interests

Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, Trustworthy AI, Time-Series Forecasting, Applied Data Science

Education

BRAC University Ongoing
M.Sc. in Computer Science and Engineering

BRAC University 7 May 2018 – 17 January 2023
B.Sc. in Computer Science and Engineering CGPA: 3.63/4.00
Thesis: ML Based Career Suggestive System for Informal Job Sector Considering Cognitive Skills

Academic Appointments

European University of Bangladesh 18 March 2025 – Present
Lecturer, Department of Computer Science and Engineering

- Courses taught: Object-Oriented Programming (OOP), Digital Logic Design, Software Engineering, Artificial Intelligence, Image Processing, Theory of Computing, and Computer Fundamentals.

BRAC University 4 October 2020 – 28 May 2022
Student Tutor

- Supported undergraduate learning and helped explain core computer science concepts in tutorial settings.

Professional Experience

SYMPHONY Softtech Ltd. 25 May 2024 – 18 March 2025
Software Developer

Square Informatix Limited 20 August 2023 – 15 February 2024
Programmer

Data Edge Limited 18 September 2022 – 18 December 2022
Intern

Publications and Conference Record

Journal / Preprint

- Enam Ahmed Taufika, Md Ahasanul Arafath, Abhijit Kumar Ghosh, Md Tanzim Reza, and Md Ashad Alam. **Histo-MExNet: A Unified Framework for Real-World, Cross-Magnification, and Trustworthy Breast Cancer Histopathology.** *arXiv preprint arXiv:2603.14416*, 2026. Submitted to *Biomedical Signal Processing and Control*.

Conference Publications and Presentations

- Fateha Jannat Ayrin, Mizanur Rahman, Ashifur Rahman, Vishwanath Akuthota, Abhijit Kumar Ghosh, Md Maruf Hasan Khondaker, Muntasim Ul Haque, Montaser Abdul Quader, and Md Tanzim Reza. **Multi-Agent Phishing Detection and Deletion via Small VLM and LLM Reasoning.** Accepted and presented at the *2026 IEEE International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2026)*, held on 29–31 January 2026 at Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh. Paper ID: 383. Proceedings published in IEEE Xplore.
- Anika Islam, Md Ahasanul Arafath, Abhijit Kumar Ghosh, Tawsif Mustasin Rafin, Turjoy Das, Simantha Saha, Fateha Jannat Ayrin, and Mizanur Rahman. **Tea Leaf Disease Classification and Out-of-Distribution Detection Using Deep Learning.** Presented at the *2025 IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE 2025)*, held on 21–22 December 2025 at Cox's Bazar, Bangladesh.

- Md Ahasanul Arafath, Md Tanzim Reza, Md. Hasan Moon, Simantha Saha, Sadia Shobnom, Md Rony Ahmed, and Abhijit Kumar Ghosh. **Comparative Multivariate Time Series Forecasting of IMU Based Gait Data for Parkinson's Disease Using VAR, LSTM, and GRU Models.** Accepted and presented at the *International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, held on 12–13 December 2025 at Daffodil International University, Dhaka, Bangladesh. Paper ID: 10382.
- Md Shahriar Sajid, Abhijit Kumar Ghosh, and Fariha Nusrat. **PaperNet: Efficient Temporal Convolutions and Channel Residual Attention for EEG Epilepsy Detection.** Accepted and presented at the *International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, held on 12–13 December 2025 at Daffodil International University, Dhaka, Bangladesh. Paper ID: 10497.
- Fateha Jannat Ayrin, Md Mizanur Rahman, Mozdaher Abdul Quader, Vishwanath Akuthota, Mehedi Hasan Jony, Md Maruf Hasan Khondaker, Nur A Shawal, Montaser Abdul Quader, and Abhijit Kumar Ghosh. **Consensus with LLMs: CoT-Arbitrated Byzantine Fault Tolerance in Network Intrusion Detection.** Accepted and presented at the *International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, held on 12–13 December 2025 at Daffodil International University, Dhaka, Bangladesh. Paper ID: 10791.
- Abhijit Kumar Ghosh, Chowdhury Sajadul Islam, Mohammad Azim, Fariha Nusrat, Md. Ahasanul Arafath, and Md Tanzim Reza. **Swin Transformer-Based Multi-Class Classification of Gallbladder Diseases: A Comparative Study with Lightweight CNNs.** Presented at the *2nd Undergraduate Conference on Intelligent Computing and Systems (UCICS 2026)*, held on 17–18 January 2026 at Varendra University, Rajshahi, Bangladesh. Paper ID: 18.

Ongoing Conference Submissions

- **PDFGuard: Multi-Layer Adversarial Protection Against AI-Based Automated Peer Review.** Submitted to the *IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*.
- **ET-AMAF: Adaptive Multi-View Transformer for Banknote Detection.** Submitted to the *IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*.
- **Reasoning Vision-Language Models for Multimodal Political Discourse and Bot Detection.** Under review at the *2026 International Conference on Engineering and Frontier Technologies*.

Academic Service

- Reviewer, Deep Learning IndabaX Nigeria 2026. Served as a peer reviewer for an international AI/ML conference under the Deep Learning Indaba initiative, contributing to capacity building in African AI research. Reviewed submissions in **Computer Vision & Multimodal AI** (Primary), **AI for Health & Biomedical Applications**, and **Distributed, Federated & Privacy-Preserving AI**, evaluating originality, technical rigor, and real-world applicability, particularly in resource-constrained environments.

Selected Projects

AI-Based Grade Prediction System

- Built a machine learning-based student performance prediction system using academic indicators such as quizzes, attendance, assignments, and participation.
- Used Python, scikit-learn, and SQL Server to analyze historical student data and identify at-risk learners.

Bangla Language Developing System

- Proposed a dedicated Bangla language learning and assessment platform aimed at supporting education and job-sector examinations in Bangladesh.

Fire Alarm System in Farm

- Developed an Arduino Nano-based fire alarm system for early detection of fire hazards in farm environments.

Honors and Academic Recognition

- Vice Chancellor's List Honoree for outstanding academic performance in 3 semesters.
- Dean's List Honoree for academic excellence in 1 semester.

Certifications

- Full Stack ASP.NET Core, PencilBox Training Institute (2023)
- Programming for Everybody (Getting Started with Python), University of Michigan (2020)
- Python Data Structures, University of Michigan (2020)
- Introduction to HTML5, University of Michigan (2020)

- Programming Fundamentals, Duke University (2020)
- Presentation Skill, BRAC University (2018)

Technical Skills

Programming and Scripting: Python, Java, C++, C#

Research Specializations: Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, Trustworthy AI, Multimodal AI

Research and Analytical Competencies: Data Preprocessing, Feature Engineering, Model Development, Experimental Design, Performance Evaluation, Statistical Analysis

Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV

Databases: MySQL, Oracle

Web and Software Technologies: HTML, CSS, JavaScript, Apache Web Server

Academic and Development Tools: LaTeX, Git, Jupyter Notebook, Google Colab

Languages: Bangla (Fluent), English (Fluent), Chinese (Professional Proficiency)

References

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